

Mohit Srivastav

Curriculum Vitae

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Ph.D. Candidate at Johns Hopkins University

Education

2022

Ph.D. Candidate, Johns Hopkins University (JHU)

Advisor: Professor Andrei Gritsan

2018–22

B.S. Physics, B.A. Computer Science, University of Virginia (UVA)

Highest Distinction in Physics, High Distinction in Computer Science

Selected Publications

CMS Collaboration. Determination of the spin and parity of all-charm tetraquarks. *Nature*, 648(8092):58–63, December 2025.

CMS Collaboration. Measurement of the higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV. *Physical Review D*, 111(9), May 2025.

CMS Collaboration. Combined measurements and interpretations of higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ tev. *arXiv*, 2026.

Jeffrey Davis, Andrei V. Gritsan, Lucas S. Mandacaru Guerra, Lucas Kang, Michalis Panagiotou, Jeffrey Roskes, and Mohit Srivastav. Maximizing returns: Optimizing experimental observables at the Lhc. *arXiv*, 2026.

Analysis Work

2023

Off-shell Higgs Boson, *CMS Joint 4-Lepton Study Team (CJLST)*

- Expert in off-shell analyses on CMS, working towards SM EFT interpretation of the off-shell Higgs boson
- Leading efforts to calculate new k-factors for higher-order behavior in multidimensional space, as presented in the LHC Higgs Working Group
- Member of the local organizing committee for the 2026 LHC Physics Center Effective Field Theory (EFT) at JHU
- Produced leading constraints on the mass and width of the Higgs Boson as well as anomalous Higgs Boson couplings with the CMS detector [2]
- Collaborated to include an off-shell Higgs analysis into the CMS Higgs Grand Combination for the first time [3]

2022–25

Spin-Parity Analysis of a Tetraquark Candidate, *CMS B-Physics*

- Found evidence for the spin and parity of 3 low-mass resonances which constitute a candidate for a tetraquark [1]
- Utilized techniques pioneered in the Higgs sector and developed them for use in B-physics

2020–22

Optimizing Cuts for Experimental Dead-time, *Mu2e Collaboration*

- Utilized machine learning to minimize experimental deadtime at high beam intensities, and studied the aging of test-stand di-counters for particle detection culminating in a [Thesis](#) in Computer Science

Technical Work

2025

Lead Developer/Maintainer, *MiLoMerge* Package

- Developer/maintainer for the [MiLoMerge](#), package, designed to merge bins optimally to reduce separability loss between N distributions [4].
- Writing [documentation](#) and [unit-tests](#) for MiLoMerge in-line with Scikit-HEP requirements
- Version 1.0.0 released on January 15, 2026

2023–24

Parton Showering and Filtering, *CJLST*

- Worked in conjunction with experts at Fermilab to establish better parton showering for off-shell Higgs Boson simulation to better match data
- Produced a better filter for the 4-lepton channel to be used in off-shell Higgs Boson simulation

2022

JHUGen/JHUGen-MELA Development, *Johns Hopkins University*

- Developer for the [JHUGen Package](#), which includes a Monte Carlo generator as well as a reweighting scheme for experimental high-energy physics, with the latest version (v7.5.7) released Dec. 20, 2024
- Created new Python bindings using PyBind11 to release a version of JHUGen-MELA in Python
- Writing [documentation](#) for the JHUGen-MELA package utilizing Doxygen

2022

CMS Tracker Alignment, *Tracker Alignment Group*

- Utilize survey constraint data to provide initial constraints for the High-Luminosity LHC tracker upgrade alongside teams at Cornell and Fermilab
- Provide alignments for the CMS tracker and validate performance over time

Public Talks & Posters

2026

Binning Bonanza: Metric Design for Quantum Observables and Efficient Dimensionality Reduction

[Pheno 2026](#) – Pittsburgh, PA

2026

Status of k-factors for Run-3 analyses

[LHC Higgs Cross Section Working Group](#) – LHC Higgs Working Group

2025

Maximizing Impact: The Quest for Optimal Observables in Experiments

[JHU Experimental Particle Physics Seminar](#) – Johns Hopkins University, MD

2025

Blasting Off(-shell): $H^* \rightarrow ZZ \rightarrow 4\ell$ for Precision Higgs boson Measurements at CMS

[Fermilab LHC Physics Center Forum](#) – Fermi National Accelerator Laboratory, IL

2025

Measurement of the Higgs boson width using the ZZ final state

[Fermilab Annual Users and Affiliates Meeting](#) – Fermi National Accelerator Laboratory, IL

2025

JHUGen-MELA Tutorial for Higgs SM EFT

[LPC EFT Workshop](#) – Fermi National Accelerator Laboratory, IL

2024

JHUGen-MELA Tutorial for Higgs SM EFT

[LPC EFT Workshop](#) – University of Notre Dame, IN

2024

JHUGen-MELA Tutorial for Off-shell Higgs Study

[LPC Off-shell Higgs Boson Workshop](#) – Fermi National Accelerator Laboratory, IL

2023

Analysis of Hadronic Resonance Structures in the $J/\psi J/\psi$ Invariant Mass Spectrum

[American Physical Society April Meeting](#) – Minneapolis, MN

2022

Improvements to Cosmic Muon Identification Using Machine Learning

[Virginia Space Grant Consortium Research Symposium](#) – Norfolk, VA

2021

Improvements to Cosmic Muon Identification Using Machine Learning

[APS Meeting of the Division of Particles and Fields](#) – Florida State University, FL

Awards & Honors

	Paul G. Agnew Teaching Award (May 2026)
	Krieger School of Arts and Sciences Excellence in Teaching Award (April 2026)
	Phenomenology 2026 Symposium Student Travel Award (April 2026)
	LPC Guest & Visitor Award (May 2025)
	E.J. Rhee Flair in Teaching Award (May 2025)
	Kupperman Family Department Travel Grant (May 2025)
	Universities Research Association Visiting Scholars Program (Mar. 2025)
	CMS IRIS-HEP Analysis Software Training Travel Grant (Mar. 2024)
	URA P5 Fermilab Travel Grant (Dec. 2023)
	Mitchell Undergraduate Research Scholarship (Jun. 2021)
	Virginia Space Grant Consortium Research Scholarship (Apr. 2021)
	SULI Argonne Lab (Jan. 2021) Offer extended; did not accept
	DAAD Rise Research Fellowship (Mar. 2020) Offer extended; cancelled due to COVID-19
	SULI Jefferson Lab (Jan. 2020) Offer extended; cancelled due to COVID-19

Selected Teaching Experience

	Head TA , <i>Classical Mechanics for Physics Majors</i> , Fall 2024, 2025 Head TA for Physics 1 for Physics Majors
	TA , <i>General Physics Lab</i> , Fall 2023 – Spring 2024 TA for General Physics Lab 1 & 2
	TA , <i>Computer Algorithms</i> , Summer 2020 – Spring 2021 TA for CS 4102
	TA/Head TA , <i>Introductory Computer Science in Python</i> , Spring 2019 – Spring 2022 TA and Head TA for CS 1110/1111

Physics Outreach

	Johns Hopkins Physics Fair (April 2026) Part of the LHC Exhibit talking showcasing the cloud chamber and muon detector
	Fairfax Science and Engineering Fair Category Judge (March 2026) Judging high school students at the regional science fair in Fairfax, Virginia
	Johns Hopkins Physics Fair Volunteer (April 2025) Performed the “Ping-pong cannon” demo for local families and talked about shockwaves/air pressure

Workshop & Conference Organizing

	LPC EFT Workshop Local Organizing Committee On the local organizing committee for the LPC EFT workshop held at Johns Hopkins
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Other Experience

2023-24

Fitness Monitor, *Ralph O'Connor Rec Center*, JHU

Helped manage facilities, play music, and interact/help patrons at the gym

2020-22

Writer, *The Yellow Journal*, UVA

Collaboratively wrote articles and headlines for the Yellow Journal, the University of Virginia's satirical publication

2019-21

Treasurer, *International Relations Organization*, UVA

Managed taxes and finances on the order of \$30k for the 501c3 International Relations Organization at the University of Virginia